HEXBOARD SEQUENCER MODE DETAILED MANUAL

1. Purpose

Sequencer mode turns the HexBoard into a separate pattern sequencer without replacing Keyboard mode. Keyboard mode and Sequencer mode are deliberately split, so you can return to the normal board behavior at any time.

1. Current Sequencer Design

- Track count: 1 track
- Step count available: 32 steps
- Active playback length: 1 to 32 steps
- Notes per step: up to 4
- Note storage format: tuning-relative pitch steps
- Length per step: stored per step
- Velocity per step: stored per step
- Probability per step: stored per step
- Output type: MIDI or onboard synth
- Storage: onboard flash files in /Sequences

1. Entering And Leaving Sequencer Mode

From the main menu, select Sequencer.

To leave Sequencer mode, open the Sequencer menu and select Keyboard.

1. Physical Layout In Sequencer Mode

4.1 Step keys

The step grid uses the first 8 playable keys from the first 4 rows:

- Row 1: steps 1-8
- Row 2: steps 9-16
- Row 3: steps 17-24
- Row 4: steps 25-32

4.2 Dedicated sequencer keys

- Top-right key: Play/Stop
- Second-row 9th key: step overview page button
- Second-row far-right key: blue action key

- Third-row far-right key: step tools picker
- Hold the top-right key for 2 seconds to show the performance monitor while held
- The overview and tools buttons stay lit white while idle and become brighter white while their overlays are active

4.3 Lower keyboard area

The lower part of the board remains playable in Sequencer mode. These keys:

- audition notes at any time
- assign or remove notes from the selected step when a step is selected
- if the shared `DisplayNotes` setting is enabled, they can also drive the `Now Playing` overlay when no step is selected

1. Selecting And Editing Steps

5.1 Selecting a step

- Press a step once to select it.
- Press the same step again to deselect it.
- The selected step blinks so it is easy to identify.

5.2 What happens when a step is selected

When a step is selected:

- the Edit overlay appears
- the Edit overlay shows the selected step number, a compact `L ...% V ... P ...%` line, and the notes
- note keys add or remove notes from that step immediately
- the step stores notes sorted from lowest to highest
- the step can be length-edited immediately with the encoder

5.3 Empty steps

- Empty steps are dark when unselected.
- A selected empty step still blinks so it is visible.

1. Entering Notes

6.1 Live note entry

With no step selected:

- pressing lower keyboard notes auditions them only

- if the shared `DisplayNotes` setting is enabled, these manually played notes can appear in the `Now Playing` overlay

With a step selected:

- pressing a note key auditions that note
- the note is also added to the step if it was not already there
- pressing the same note again removes it from the step
- this step-edit workflow does not drive the `Now Playing` overlay

Sequencer note entry uses the current Keyboard tuning and transpose rather than storing raw MIDI note numbers.

6.2 Chords

- Each step can hold up to 4 notes.
- Notes are displayed in ascending order.
- In 12 EDO, note labels use normal note names like `C3`.
- In other tunings, note labels use `step.octave` labels like `7.4`.
- Playback sends all notes in the step together.
- Sequencer playback itself does not drive the `Now Playing` overlay.

6.3 Tap Preview

The Tap Preview menu item controls whether tapping a step previews its stored note or chord when you select it.

- Off: tapping a step only selects it
- On: tapping a step selects it and previews its contents

1. Note Length

7.1 How to change length

- Select a step
- Turn the encoder for quick length changes
- Open the step tools picker and choose Len for exact length entry

The available length values are:

- 0%
- 25%
- 50%

- 75%
- 100%
- 150%
- 200%
- 250%
- 300%
- 350%
- 400%
- 500%
- 600%
- 700%
- 800%
- 900%
- 1000%

7.2 What the values mean

Length is stored as a percentage of the step duration.

- 100% means one full step
- values below 100% shorten the note
- values above 100% let the note continue across later steps
- 0% means the step will not sound

7.3 Exact length entry

Choosing Len from the step tools picker opens the exact length screen.

In exact length mode:

- number keys are used to enter the value directly
- the OLED shows the current saved value and a separate new-entry field
- typing digits fills the `New:` field without overwriting the `Current:` field
- up to 4 digits are allowed
- the exact-length keypad moves to the lower part of the board so the top 4 sequencer step rows remain visible
- exact-length keypad keys are lit blue at the sequencer medium light level
- <- removes the previous digit and is on the eighth-row first key

- CANCEL exits exact length entry without saving and uses the same lower cancel area as the tools picker
- pressing the encoder saves the `New:` value when one was entered, or keeps the current value if `New:` is blank, then returns to the tools picker

The exact length screen shows this keypad-style layout:

- Current: 0200
- New: 0250
- 0 1 2 3 4
- 5 6 7 8 9
- <-
- CANCEL

1. Transport And Playback

8.1 Play/Stop key

The top-right key toggles transport:

- red when stopped
- green when playing
- if a step is selected, a short Play/Stop press deselects it before toggling transport

Press and hold the Play/Stop key for about 2 seconds to open the performance monitor. The monitor stays visible only while the key is held and closes when the key is released.

The performance monitor shows:

- CPU load as an approximate percentage based on the audio poll interrupt load
- heap memory used versus total
- LittleFS storage used versus total

8.2 Tempo

Tempo is set in Playback Settings. When Clock Source is Internal, the sequencer treats each step as a 16th-note step at the current BPM.

8.3 Clock Source

Clock Source is in Playback Settings > MIDI Sync.

- Internal: the sequencer uses its own Tempo setting as the active step clock
- External MIDI: incoming MIDI clock drives the sequencer timing

When Clock Source is External MIDI:

- incoming MIDI Start starts the sequencer from the beginning
- incoming MIDI Stop stops the sequencer
- local tempo remains stored, but external MIDI clock is the active timing source while running

8.4 Clock Send And Transport Send

Playback Settings > MIDI Sync also includes:

- Send Clock
- Send Transport

Current behavior:

- Send Clock only applies when Clock Source is Internal
- when Send Clock is On, the sequencer sends MIDI clock while transport plays
- Send Transport only applies when Clock Source is Internal
- when Send Transport is On, local Play/Stop sends MIDI Start and Stop

8.5 Steps

The Steps menu item decides how many steps are active in playback:

- minimum 1
- maximum 32

All 32 steps still remain editable even if the active playback count is lower. Step LEDs beyond the active playback count stay dark, even if those later steps still contain notes.

8.6 Direction

Available direction modes:

- Forward: step 1 to end, then loop
- Backward: end to step 1, then loop
- Ping-Pong: runs forward to the end, then reverses back
- Random: picks any step in the active range each tick
- Brownian: moves only to neighboring steps
- Drunk: may stay still or move one step left or right

1. Output Mode

Play Type controls where the sequencer sends notes:

- MIDI: standard external MIDI behavior
- OB Synth: sends the sequence into the onboard synth engine

OB Synth uses the shared synth settings available from Synth Options.

Sequencer playback resolves stored step pitches through the current Keyboard tuning and transpose at playback time, so the same saved pattern can follow different tunings later without changing the file format again.

1. Undo, Clear, And Step Tools

10.1 Blue action key

The blue action key on the far right of the second row has two jobs:

- short press: undo the currently selected step back to the state it had when you selected it, including notes, length, velocity, probability, and Tie
- hold for about 1 second: clear the selected step

After a clear, the screen reminds you that the blue key can undo the erase.

10.2 Step tools picker

Press the far-right button on the third row while a step is selected to open the step tools picker.

Current tool choices:

- Len
- Vel
- Oct+
- Oct-
- Prob
- Tie
- Copy
- Cancel/Finished

The picker screen shows:

- Tools #step number
- the selected step's compact `L ...% V ... P ...%` line
- the selected step's current notes
- Len Vel Oct+ Oct-
- Prob Tie Copy

- Cancel/Finished

Current behavior:

- The tool buttons live in the lower part of the board so the 4 sequencer step rows stay visible at the top while the picker is open.
- Tool selection keys are lit blue at the sequencer medium light level.
- While the picker is open, tapping another step pad switches the selected step and keeps the picker open on that step.
- Len: opens exact length entry for the selected step
- Vel: opens exact velocity entry for the selected step
- Prob: opens exact probability entry for the selected step
- Oct+: transposes all notes in the selected step up 1 current tuning cycle and stays in the picker so you can apply another tool immediately
- Oct-: transposes all notes in the selected step down 1 current tuning cycle and stays in the picker so you can apply another tool immediately
- Oct+ and Oct- are limited to displayed octave range 0 through 9
- If Oct+ or Oct- hits that range limit, the range message closes back to the tools picker.
- While that tools range message is visible, further key presses do not trigger normal keyboard actions and only extend the message timeout.
- Tie: toggles the selected step's Tie flag and stays in the picker
- Copy: opens destination selection, then copies notes, length, velocity, and probability, and Tie into the tapped destination step, then reopens the tools picker on that destination step for further editing
- Cancel/Finished: exits the tool picker when you are done
- If the selected step is tied, the note row shows T instead of note labels.

10.3 Step velocity

Each step stores its own velocity value.

- range: 0-127
- default: 96

This follows normal MIDI note-on velocity style behavior.

How to edit velocity:

- select a step
- open the step tools picker
- choose Vel

The exact velocity screen uses encoder entry:

- turn the encoder to move through coarse musical values
- the values are: 0, 5, 10, 15, ... 125, 127
- pressing the encoder saves and returns to the tools picker

Velocity affects:

- sequencer playback
- Tap Preview step previews
- live note audition while a step is selected
- MIDI output velocity
- onboard synth preview/play loudness approximation

10.4 Step probability

Each step stores its own playback probability value.

- range: 0-100%
- default: 100%

How to edit probability:

- select a step
- open the step tools picker
- choose Prob

The exact probability screen uses encoder entry:

- turn the encoder to move through: 0, 5, 10, 15, ... 100
- pressing the encoder saves and returns to the tools picker

Probability affects:

- sequencer playback chance for that step
- both MIDI and OB Synth playback modes

Probability does not affect:

- Tap Preview step previews
- manual note audition while entering notes

10.5 Tie

Tie is a saved per-step continuation flag for transport playback.

How to edit Tie:

- select a step
- open the step tools picker
- choose Tie

Tie behavior:

- a tied step does not send a new note-on during transport playback
- the tied step continues the nearest earlier eligible non-tied step from the same pattern pass
- the lower playable keyboard keeps the carried note lit for the full held duration while that tied continuation remains active
- the tied step ignores its own stored note, length, velocity, and probability values during transport playback
- if no valid earlier source note is active when the tied step is reached, the tied step is silent
- Tie does not wrap from the end of the pattern back to step 1 in v1

Display behavior:

- tied steps show T in note-oriented overlays instead of note labels
- the 8-step overview pages also show T by itself for tied steps

Storage and editing behavior:

- Tie is stored per step in the sequence file
- Tie is included in step undo
- Tie is included when Copy copies one step to another
- the step's stored notes remain saved internally even while Tie is on
- Tap Preview and manual step preview stay silent on tied steps

1. Overview Screen

Press the second-row 9th key to open the note overview.

- It shows 8 steps per page.
- Press the same key again to move to the next page.
- Pages cycle through the 32-step pattern.
- Empty steps show as underscores.
- Tied steps show T.

The overview is meant as a quick readout of what is in the pattern without entering edit mode for each step.

1. Menu Structure

When a menu page is open, hold the bottom command button below the encoder to temporarily make the top two command buttons act like one encoder step up or down. Releasing that bottom button returns them to their normal non-menu behavior. While editing a value, the top shortcut button increases that value and the lower shortcut button decreases it.

Main Sequencer menu:

- Keyboard
- Playback Settings
- Synth Options
- File Management
- New
- Save
- Save New
- Load
- Revert
- Seq Lights
- Color Options
- Tap Preview
- Update Firmware

Playback Settings submenu:

- Steps
- Direction
- Tempo
- Play Type
- MIDI Sync

MIDI Sync submenu:

- Clock Source
- Send Clock
- Send Transport

File Management submenu:

- Rename File
- Rename Folder
- Create Folder
- Delete File
- Delete Folder
- USB Backup

Color Options submenu:

- Color Mode
- Brightness
- Animation
- Rest Bright
- Dim Bright

Seq Lights submenu:

- Accent Every
- Step Color
- Step Hue

These are the same shared color settings used in Keyboard mode.

- Animation also affects the lower playable keyboard note lights that Sequencer transport playback triggers for currently sounding notes.
- Rest Bright controls normal resting note brightness and regular sequencer step-light brightness.
- Dim Bright controls out-of-scale note brightness only while Scale Lock is Off.
- When Scale Lock is On, out-of-scale notes are off instead of dimmed.

Seq Lights controls:

- Accent Every emphasizes every Nth step, starting from step 1.
- Accent Every choices are Off, 2, 3, 4, 5, 6, 7, and 8.
- Accent Every uses brightness changes only and does not change hue.
- The default Accent Every value is 4.
- Step Color chooses between: Note and Regular.
- The default Step Color value is Regular.

- Sequencer step lights use this brightness ladder: off, low, medium, high, highest.
- The current step mapping is: empty step without accent = off empty step with accent = medium step with note = medium accented step with note = high selected step with note = high playing step with note = highest
- A tied step uses the same programmed-step brightness behavior as a step with a note.
- During transport playback, the lower playable keyboard lights the currently sounding notes for their full held duration, including tied continuations.
- In Note mode, programmed steps use note-based colors with that brightness ladder.
- In Regular mode, programmed steps use the selected Step Hue with that brightness ladder.
- Empty unselected steps without accent stay off in both modes.
- A selected or playing empty step lights at the highest level so selection and playback are still visible.
- Step Hue is available only while Step Color is set to Regular.
- Step Hue choices are: Red, Orange, Yellow, Lime, Green, Teal, Cyan, Lt Blue, Blue, Indigo, Purple, Magenta, and Pink.
- The default Step Hue value is Indigo.
- Step Hue previews live while you scroll through its choices, and saves when you confirm the selection.

1. New, Saving, Loading, And Reverting

13.1 New

New clears the sequencer to a blank untitled sequence.

It also clears the current loaded file path, so the title returns to Sequencer until you save the new sequence.

13.2 Save

Save writes the current sequence back to the currently loaded file.

If no current file exists yet, Save opens folder selection first, then sequence naming, and saves after you choose a location and enter a name.

13.3 Save New

Save New opens the sequence browser so you can choose a folder, then name a new sequence file in that location.

13.4 Load

Load opens the browser and loads a saved sequence file.

13.5 Revert

Revert discards unsaved edits and reloads the last saved version of the current file.

If no current file exists:

- it may fall back to the old legacy file if present
- otherwise it resets to a blank/default sequence

1. File And Folder Organization

Sequences are stored in LittleFS under:

/Sequences

The current file path is remembered, so the last loaded sequence can come back on startup.

Files use the .hbseq extension internally.

The visible sequence name is the file name without the extension.

Browser behavior:

- the browser shows up to 8 file or folder rows at a time
- `^ more ^` means there are earlier pages
- `v more v` means there are later pages
- turning the encoder past the first or last real file row changes page
- USB Backup uses the same LittleFS storage, but transfers files through a dedicated USB serial session instead of mounting the board as a drive

1. Renaming, Creating, And Deleting Files

15.1 Rename File

- browse to a file
- select it
- naming mode opens with the current file name filled in

15.2 Rename Folder

- browse folders
- enter the folder you want
- choose Rename This Folder

15.3 Create Folder

- browse to the parent folder
- choose Create Here
- naming mode opens

15.4 Delete File

- browse to a file
- select it to delete it

15.5 Delete Folder

- browse to the folder
- enter it
- choose Delete This Folder

Folder deletion is recursive, so deleting a folder removes its contents.

15.6 USB Backup

- open File Management, then choose USB Backup
- choose Start Session
- the transport is stopped and the backup page shows live status lines
- run the host backup tool on the connected computer
- if you try to leave the USB Backup page while a session is active, the sequencer warns that leaving the page closes USB Backup and asks whether you want to continue
- if you choose Stop Session while a session is active, the sequencer warns that stopping USB Backup ends any current transfer and asks whether you want to continue
- while USB Backup is active, the hex buttons are temporarily disabled and their LEDs are blanked so you do not accidentally trigger other sequencer actions; the encoder still works for menu navigation and confirmation
- choose Stop Session when you are done

Computer-side tools:

- `scripts/hexboard_backup.py` provides the command-line workflow
- `scripts/hexboard_backup_gui.py` provides a desktop GUI on macOS and Windows
- for skilled users, the simplest setup is to install Python 3, install `pyserial`, and then launch the GUI locally
- the repo root includes simple double-click launcher files: `Launch HexBoard Backup.command` for macOS and `Launch HexBoard Backup.bat` for Windows

Available host-side operations include:

- list the remote /Sequences tree
- back up or restore the full /Sequences tree

- pull or push one sequence file
- pull or push one sequence folder
- rename one remote file or folder
- delete one remote file or folder
- the desktop GUI uses four action rows overall: one row beside the serial port dropdown for `Refresh Ports` and `Refresh Remote`, then three rows below for the remaining actions
- the lower rows group: full backup actions, download and upload actions, then rename and delete actions
- use the GUI buttons to refresh ports, refresh the remote tree, back up all, restore a backup, download the selected entry, upload a file, upload a folder, rename a selected file, rename a selected folder, or delete the selected entry
- GUI renaming only allows simple sequencer-style names: letters, numbers, spaces, and –
- GUI renaming rejects periods and limits names to 20 characters so renamed files and folders stay aligned with the on-device naming workflow
- if a rename is not allowed, the GUI shows the full rename rules, points out the broken rule(s), and then reopens the rename prompt with the previous entry still there so you can fix it quickly
- in the remote tree, Shift-click can select multiple items for a shared delete action
- GUI Upload Folder always merges into the target folder and does not delete the existing remote folder first
- during Upload Folder, each conflicting remote file is shown one at a time so you can overwrite it, skip it, or cancel the folder upload
- Upload File warns before replacing an existing remote file, whether that file was selected directly or is already inside the selected target folder
- that warning also tells you to select a folder first if you want to copy the file into that folder instead of replacing a selected file
- when multiple items are selected, single-target actions such as Download Selected, Upload File, Upload Folder, Rename File, and Rename Folder are disabled until the selection returns to one item
- the GUI remembers the last local folder used for Upload File and Upload Folder, and it also remembers the last backup folder used for Restore Backup. Those locations return after the app is closed and reopened
- the top GUI status line shows active operations, including Refresh Remote
- the `Size Bytes` column shows raw file sizes in bytes
- users who already have Python 3 and `pyserial` installed can launch the GUI by double-clicking the platform launcher file in the repo root instead of typing the launch command each time

While the USB backup session is active:

- Save is blocked
- Revert is blocked
- hex-button presses and releases are ignored
- the sequencer button LEDs are blanked
- browser-based Load, Save New, Rename, Create Folder, and Delete workflows are blocked until the session is stopped
- the board does not present /Sequences as a normal computer drive

1. Naming Mode

Naming mode is used for:

- Save New
- Create Folder
- Rename File
- Rename Folder

16.1 Character layout

The on-screen keyboard layout is:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z SPC - 1 2 3 4 5 <- 6 7 8 9 0 CANCEL

16.2 Naming controls

- letter and symbol keys insert characters
- SPC inserts a blank space
-
  - inserts a hyphen
- 0 through 9 insert those digits
- <- deletes the previous character
- CANCEL exits naming mode without saving
- encoder press saves the name
- the physical CANCEL key is placed lower on the board to reduce accidental presses

16.3 Naming visuals

While naming is active:

- valid input keys are lit blue at the sequencer medium light level
- non-input keys are dark
- the display shows a blinking cursor in the name line

Maximum name length is 20 characters.

1. Performance Monitor

The sequencer has a temporary performance monitor overlay.
How to open it:

- press and hold the Play/Stop key for about 2 seconds

How it behaves:

- it appears while the key is still held
- it closes immediately when the key is released
- it does not replace the normal sequencer screen permanently

What it shows:

- AudioEng: approximate audio engine load percentage
- Mem: heap used versus total heap
- FS: LittleFS storage used versus total storage
- MIDI Q: current incoming MIDI backlog in pending bytes
- Drop: backlog episodes that hit the receive soft limit
- Late: backlog episodes that stayed pending long enough to count as delayed

While active sequencer overlays are open, the display is kept awake so edit screens do not fall asleep mid-entry.

1. Header Title And Unsaved Changes

At the top of the Sequencer menu:

- if no file is loaded, the title is: Sequencer
- if a file is loaded, the title is: Seq-FILENAME
- if there are unsaved edits, the title is: Seq-*FILENAME

The star marks the sequence as dirty, meaning the current state differs from the last saved version.

1. What Is Stored In A Sequence File

The current save file stores:

- tempo
- active step count

- play type
- direction
- the tuning-relative pitch-step values in each step
- the gate/length value for each step
- the velocity value for each step
- the probability value for each step
- the Tie flag for each step

Tap Preview is a general sequencer setting and is not stored per sequence file. It is saved with the board profile and Auto-Save settings system.

Clock Source, Send Clock, and Send Transport are general sequencer settings and are not stored per sequence file. They are saved with the board profile and Auto-Save settings system.

Seq Lights settings are general sequencer settings and are not stored per sequence file. They are saved with the board profile and Auto-Save settings system.

Color Options settings are shared with Keyboard mode and are also saved with the board profile and Auto-Save settings system.

Current sequence files are saved as `version=3` with `noteFormat=stepsFromC`. Older files that do not contain Tie fields load with Tie off on every step.

1. Current Limits

- 1 track
- 32 steps
- 4 notes maximum per step
- naming uses uppercase letters plus a small set of extra characters
- backup and restore are available over a dedicated USB serial session
- there is still no direct computer-drive mode for the sequence files

1. Practical Workflow Example
2. Enter Sequencer mode.
3. Press a step to select it.
4. Play notes on the lower keyboard area to build the step.
5. Turn the encoder while the step is selected to set note length quickly.
6. Repeat for other steps.
7. Press the top-right key to start playback.
8. Adjust Tempo, Steps, Direction, or Play Type from Playback Settings, and Tap Preview from the main Sequencer menu, as needed.

9. Choose Save or Save New to store the sequence.
10. Quick Troubleshooting

If a step does not sound:

- make sure the step contains notes
- check that length is not 0%
- check whether Play Type is set to the output you expect

If edits disappear:

- use Save to write them to flash
- otherwise Revert or power loss can discard unsaved changes

If you cannot find a sequence:

- use Load and browse the /Sequences folders
- remember that the visible name comes from the file name